MIL-E-1/988F
15 April 1976
SUPERSEDING
MIL-E-1/988E
15 January 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VOLTAGE REGULATOR

TYPE 7615

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Corona discharge

See figure 1

Mounting position: Any

Weight: 0.23 ounce (6.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ez	Eb	Ib	TA
Unit:	Vdc	Vdc	µAdc	°C
Maximum:	730	690	55	+50
Minimum:	---	670	5	-20
TEST CONDITIONS:	---	---	---	---

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (total darkness)	3		0.65	II	Ez	---	730	Vdc
3337	Voltage drop	4	Ebb/Ib = 20 µAdc	0.65	II	Eb	670	685	Vdc
3335	Regulation	4,5	Ib = 5 to 55 µAdc	0.65	II	ΔEb	---	4	Vdc
----	Peak current	2	Ebb/Ib = 150 µAdc	0.65	II	---	---	---	---
----	Holding period end point:								
3347	Ionization voltage	3,7		0.65	II	Ez	---	730	Vdc
----	Change in voltage drop	4,7				ΔEb(1)	---	±3	Vdc
3335	Regulation	4,7				ΔEb	---	4	Vdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
3305	Leakage current	-	Eb = 500 Vdc; T = 25° ±5°C; relative humidity= 90 ±5%	---	---	LIb	---	0.5	µAdc
1026	Temperature range, operating	6	T = -20° to +50°C; Ib = 20 µAdc	---	---	---	---	---	---
	Quality conformance inspection, part 3								
----	Life test	-	Group A; Ebb/Ib = 20 ±2.5 µAdc; t = 500 hours	---	---	---	---	---	---
----	Life-test end points:								
3347	Ionization voltage	3		---	---	Ez	---	730	Vdc
3337	Voltage drop	4		---	---	Eb	670	690	Vdc
3335	Regulation	4		---	---	ΔEb	---	5	Vdc
1041	Shock	8	Hammer angle = 20°	---	---	---	---	---	---
(F)1031	Low-frequency vibration	1.8	No voltages applied; F = 25 Hz	---	---	---	---	---	---

NOTES:

1. Criterion for passing the vibration test shall be compliance after test with initial requirements for voltage drop and regulation.
2. Ebb shall be increased continously until Ib = 150 µAdc. The voltage drop shall not go below its value at Ib = 55 µAdc.
3. The tube under test shall not have been in conduction for at least 30 minutes prior to test. During this test the regulator tube shall not be exposed to external sources of radiation and shall be shielded from light. No conditioning current is permitted.
4. The tube shall be operated at the current specified for a period of at least 10 minutes prior to taking readings.
5. ΔEb maximum shall not be exceeded anywhere within the range 5 to 55 µAdc. In testing for regulation, current shall be varied continuously through the range of 5 to 55 µAdc.
6. ΔEb shall not exceed +4 Vdc in going from room temperature (25° ±5°C) to +50°c. ΔEb shall not exceed the limits of 0 to -5 volts when going from room temperature to -20°C. Ionization voltage and regulation shall remain within room temperature limits.
7. The holding period commences upon completion of all production tests. At the conclusion of the holding period the specified tests shall be performed again. Of the tubes failing the holding period end-point test, those which fail with respect to change in voltage drop only, may be retested after an additional 30-day period. ΔEb(1) is the change in voltage drop from beginning to end of the holding period.

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(F) 8. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

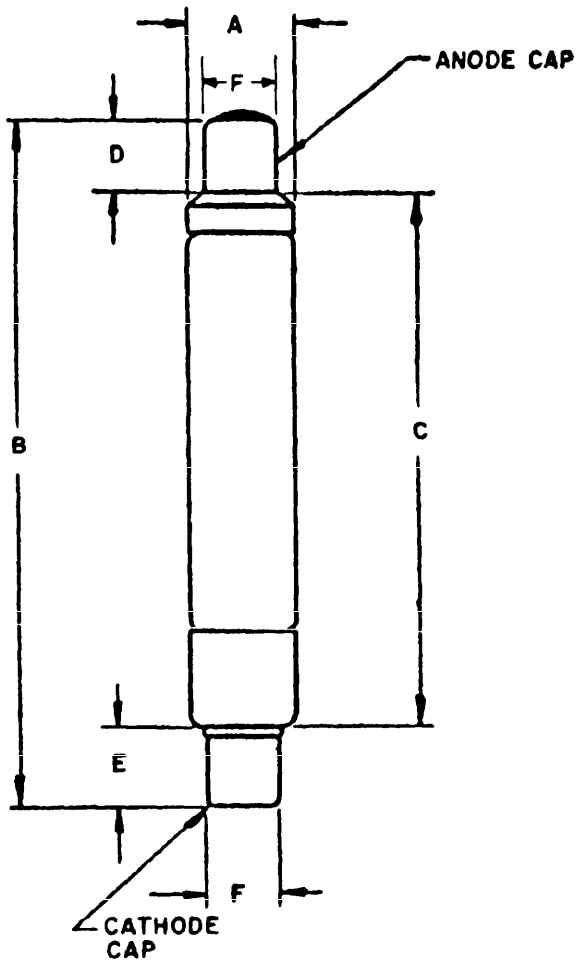
User activities:
Army - WC, MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3085)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		.395 (10.03)
B	2.125 (53.98)	2.250 (57.15)
C	1.625 (41.28)	1.750 (44.45)
Reference dimensions		
D	.250 (6.35)	
E	.250 (6.35)	
F	.250 (6.35)	

NOTE:
Mechanical configuration and materials used may vary, provided all specified dimensions are met.

FIGURE 1. Outline drawing of electron tube type 7615.

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